

Creating and Sustaining an Effective e-waste Management System

It's high time that all concerned stakeholders come together to create an effective mechanism for collection, recycling and disposal of e-waste



E-waste encompasses ever-growing range of obsolete electrical and electronic goods

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Increased consumption of electronic products has led to a substantial rise in the quantum of e-waste being generated along with rapid depletion of natural resources. This new kind of waste is posing a serious challenge in India as the country lacks environmentally safe disposal and recycling processes. There is an urgent need to evolve policies and systems to deal with the growing menace of e-waste.

E-waste and its sources

E-waste is a collective term for any discarded electronic product that contains substances known to be:

1. Toxic and fatal to humans
2. Carcinogenic, mutagenic or tera-

togenic to human or life-forms

3. Corrosive
4. Highly reactive (undergoes violent chemical reactions either by itself or when mixed with other materials)
5. Explosive or ignitable with a flash point less than 60°C

E-waste encompasses ever-growing range of obsolete electrical and electronic goods that are broadly classified as white goods comprising household appliances like refrigerators, microwave ovens, air-conditioners and washing machines; brown goods comprising television sets, audio and video players, cameras, etc; and grey goods like computers, printers, copiers, scanners and fax machines.

E-waste contains a number of toxic substances, such as lead and cadmium in cathode ray tube (CRT) monitors,

mercury in switches and flat-screen monitors, polychlorinated biphenyls (PCBs) in older capacitors and transformers, and brominated flame retardants in printed circuit boards, plastic castings, cables and polyvinyl chloride (PVC) cable insulations.

Major contributors to e-waste in India are:

Individual households. Individual households are not large contributors to waste generated by computers, but these certainly consume large quantities of consumer durables and are potential creators of waste thereof.

Business sector. Government departments, public/private enterprises and MNC offices are major producers of obsolete technology due to incompatibility of old systems to meet current application needs and requirements.

Manufacturers and retailers. The waste from manufacturers and retailers comprises defective integrated chips, motherboards, CRTs and other peripheral items produced during manufacturing and production process. It also includes electronic items and products procured in exchange from the customers.

Imports. Monitors, keyboards, CPUs, typewriters, mobile phones, projectors and PVC wires being brought in illegally add to the volume of waste being generated within the country. Lack of understanding of national policies on import and export and porous ports at both the points of origin and final destination also add to the volumes. Availability of cheap labour and weak environmental laws are largely responsible for promotion of such illegal trade.